Requirements and Acceptance Criteria

Project: Development of a Web-Based Volunteer and Event Management System for Astana IT University: Recruitment, Intelligent Matching, and Activity Coordination

Team members:

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1. Priority Levels

Priority	Meaning
P0 — Must have	Essential for a working MVP and the main user workflows.
P1 — Should have	Important for effective event coordination and user experience.
P2 — Could have	Additional functionality that can be implemented if time permits.

2. Functional Requirements

FR-01. User Registration and Authentication

Priority: P0

Requirement: The system shall allow users to register, log in, and securely access their accounts.

Acceptance criteria:

- A user can register using the required account information.
- The system validates required fields and rejects invalid input.
- The system prevents registration with an email address already associated with an account.
- A registered user can log in using valid credentials.
- Invalid credentials do not grant access and produce an appropriate error message.
- Unauthenticated users cannot access protected account pages.
- Passwords are not stored in plain text.

FR-02. User Profiles

Priority: P0

Requirement: The system shall allow volunteers to create and update profiles containing information relevant to volunteering opportunities.

Acceptance criteria:

- A volunteer can view and edit their profile.

- A profile can include skills, interests, experience, and availability.
- The system validates profile data before saving changes.
- Updated profile information is displayed after successful saving.
- The matching process uses the volunteer's current profile information.
- Users cannot modify another volunteer's profile without authorization.

FR-03. Event Creation and Management

Priority: P0

Requirement: The system shall allow authorized organizers to create, edit, publish, and manage volunteering events.

Acceptance criteria:

- An organizer can create an event with a title, description, date, time, location or meeting details, and recruitment information.
- The system validates required event fields and rejects invalid dates or times.
- An organizer can save an event as a draft before publishing it.
- Published events are visible to eligible volunteers.
- Organizers can edit or cancel events they are authorized to manage.
- Event status is displayed clearly.
- Cancelled or unpublished events cannot receive new applications.

FR-04. Event Roles and Requirements

Priority: P0

Requirement: The system shall allow organizers to define volunteer roles within an event.

Acceptance criteria:

- An organizer can create one or more roles for an event.
- Each role includes a description, required skills, scheduled time, and number of volunteer positions.
- Additional qualifications or participation requirements can be specified when necessary.
- The system validates the required fields and the number of available positions.
- Volunteers can view the requirements before applying.
- The system identifies roles that have reached their capacity.

FR-05. Volunteer Applications

Priority: P0

Requirement: The system shall allow volunteers to apply for available event roles and track their application status.

Acceptance criteria:

- A volunteer can submit an application for an open role.
- The system prevents duplicate active applications for the same role.
- Applications cannot be submitted for cancelled events, closed recruitment, or full roles.
- A volunteer can view their submitted applications and their current statuses.
- Application statuses include at least Pending, Accepted, and Rejected.
- An organizer can review an application and approve or reject it.
- The system records the updated status and makes it visible to the relevant volunteer.
- Only authorized organizers can make decisions about applications for their events.

FR-06. Intelligent Volunteer Matching

Priority: P0

Requirement: The system shall recommend volunteers for event roles based on their profile information and role requirements.

Acceptance criteria:

- The matching method compares volunteer skills with the skills required by a role.
- The method considers relevant interests and experience when this information is available.
- A volunteer who is unavailable during a role's scheduled time is excluded from eligible recommendations.
- Volunteers who do not meet mandatory role requirements are excluded from eligible recommendations.
- Eligible volunteers are ranked according to a documented matching method.
- The system displays recommendations and the main factors contributing to a match.
- The organizer can review recommendations before making assignment decisions.
- Organizers can assign volunteers without relying exclusively on the recommendation ranking.
- The matching process can be tested using predefined volunteer profiles and role requirements.

Evaluation requirement: Matching performance shall be compared with a simple baseline, such as ranking eligible volunteers by the number of required skills they possess. Evaluation shall use organizer-reviewed judgments or another documented evaluation method.

Scope limitation: The initial version will use a transparent, rule-based or weighted scoring approach unless sufficient data and implementation time justify a more advanced model. The system will not claim that recommendations are accurate without evaluation results.

FR-07. Volunteer Assignments

Priority: P0

Requirement: The system shall allow authorized organizers to assign volunteers to event roles and manage confirmed placements.

Acceptance criteria:

- An organizer can assign a volunteer to an available role.
- The system prevents assignments that exceed the role's capacity.
- The system prevents assignments that conflict with the volunteer's recorded availability.
- The system prevents conflicting assignments when their scheduled periods overlap.
- Only eligible volunteers can receive confirmed assignments.
- An assignment is visible to the volunteer and the responsible organizer.
- The system records the assigned role and its scheduled time.
- An organizer can cancel or update an assignment, with the change reflected in the relevant views.
- The system prevents unauthorized users from creating or modifying assignments.

FR-08. Scheduling and Conflict Management

Priority: P0

Requirement: The system shall provide schedules for events and volunteer assignments and identify scheduling conflicts.

Acceptance criteria:

- Volunteers can view their upcoming assignments and event schedules.
- Organizers can view assignments for their events.
- The system checks whether a proposed assignment overlaps with another confirmed assignment.
- The system rejects conflicting assignments unless an explicitly defined exception is permitted.
- Changes to event dates or times are reflected in relevant schedules.
- Volunteers can identify the date, start time, end time, and location of each assignment.
- The system displays clear validation messages when a scheduling conflict prevents an assignment.

FR-09. Task and Activity Coordination

Priority: P1

Requirement: The system shall support basic coordination of volunteer responsibilities before and during events.

Acceptance criteria:

- An organizer can provide task instructions for an event role or assignment.
- Assigned volunteers can view their responsibilities.
- Organizers can update instructions when necessary.
- Updated instructions are visible to the affected volunteers.
- The system distinguishes between an assigned role and any additional task information.
- Only authorized organizers can modify task instructions.

FR-10. Notifications and Status Updates

Priority: P1

Requirement: The system shall inform users about important application, assignment, and scheduling changes.

Acceptance criteria:

- A volunteer can identify when an application status changes.
- A volunteer can view newly confirmed, updated, or cancelled assignments.
- Relevant event or schedule changes are reflected in the user's account.
- Notifications identify the related event or role and the action required, if any.
- The system does not send duplicate notifications for the same recorded status change under normal operation.
- Notification delivery failures do not incorrectly change the underlying application or assignment status.

MVP limitation: In-app notifications are sufficient for the initial implementation. Email, SMS, or push notifications may be added if time and infrastructure permit.

FR-11. Attendance and Participation Records

Priority: P1

Requirement: The system shall allow authorized organizers to record volunteer attendance and participation.

Acceptance criteria:

- An organizer can view the volunteers assigned to an event or role.

- An authorized organizer can record attendance for an assigned volunteer.
- Attendance records include the related event, volunteer, date, and attendance status.
- The system distinguishes attendance statuses, such as Present, Absent, and Not Recorded.
- Only authorized users can create or modify attendance records.
- Updated attendance information is visible to the responsible organizer.
- The system prevents duplicate attendance records for the same volunteer and assignment.
- Attendance information can be used to generate participation summaries.

FR-12. User Roles and Permissions

Priority: P0

Requirement: The system shall enforce role-based access control for volunteers, organizers, and administrators.

Acceptance criteria:

- Volunteers can manage their own profiles, view opportunities, submit applications, and view their assignments.
- Organizers can manage events and roles they are authorized to manage, review applications, make assignments, and record attendance.
- Administrators can manage user access and oversee platform activity.
- Volunteers cannot publish events, approve applications, or modify other users' attendance records.
- Organizers cannot modify events managed by other organizers unless explicitly authorized.
- Unauthorized requests are rejected by the backend, even when submitted outside the normal user interface.
- The system checks permissions before executing protected operations.

FR-13. Search and Filtering

Priority: P1

Requirement: The system shall allow volunteers to discover relevant events and roles through search and filtering.

Acceptance criteria:

- Volunteers can browse published events and available roles.
- Users can search using event or role keywords.
- Users can filter opportunities by available criteria, such as event date, required skills, or event category.

- The system displays only opportunities that satisfy the selected filters.
- Cancelled, unpublished, and otherwise unavailable opportunities are excluded from the list of open opportunities.
- Clearing filters restores the unfiltered list of available opportunities.

FR-14. Reporting and Participation Summaries

Priority: P2

Requirement: The system should provide organizers with basic summaries of recruitment and participation.

Acceptance criteria:

- An organizer can view the number of applications received for an event.
- The system distinguishes pending, accepted, and rejected applications.
- Organizers can view the number of assigned volunteers and available positions.
- Attendance summaries can be generated from recorded participation data.
- Reports use stored application, assignment, and attendance records rather than estimated values.
- Users cannot access reports containing information outside their permissions.

3. Non-Functional Requirements

NFR-01. Security

Priority: P0

Requirement: The system shall protect account information and prevent unauthorized access to platform functionality.

Acceptance criteria:

- Passwords are stored using a suitable password-hashing algorithm.
- Protected backend operations verify authentication and authorization.
- User input is validated on the server.
- Secrets and credentials are stored outside the source code repository.
- Error responses do not expose passwords, authentication tokens, or confidential configuration values.
- Security checks cover attempts to access or modify another user's protected data.

NFR-02. Usability and Accessibility

Priority: P0

Requirement: The web interface shall be understandable and usable by student volunteers and event organizers.

Acceptance criteria:

- Navigation and labels are consistent across the main pages.
- Users can complete the main workflows without relying on undocumented instructions.
- Forms display understandable validation messages.
- Application, assignment, and attendance statuses are distinguishable.
- Interactive elements have clear labels and visible focus states.
- Core workflows are tested with representative users or through a documented usability evaluation.

NFR-03. Performance

Priority: P1

Requirement: The system shall respond to common user actions within an acceptable time under the expected MVP workload.

Acceptance criteria:

- Common operations, such as opening event lists, viewing profiles, and submitting applications, have a target response time of no more than 2 seconds under the documented test conditions.
- Matching results for a test dataset of 100 volunteers and 20 roles are generated within 3 seconds under the documented test conditions.
- Performance measurements record the test environment and dataset size.
- Performance tests identify failures or substantial deviations from the targets.

These targets are proposed engineering criteria and must be verified during implementation; they are not claims about existing system performance.

NFR-04. Reliability and Data Integrity

Priority: P0

Requirement: The system shall maintain consistent records of applications, assignments, schedules, and attendance.

Acceptance criteria:

- Successfully saved records remain available after the page is refreshed.
- Invalid operations do not leave partially updated application or assignment records.
- Role capacity and assignment-conflict rules are checked before confirming an assignment.

- Related database changes are handled consistently.
- The system displays an error when an operation cannot be completed instead of reporting false success.

NFR-05. Maintainability and Code Quality

Priority: P1

Requirement: The system shall use a maintainable structure that supports testing and future development.

Acceptance criteria:

- Frontend, backend, and database responsibilities are separated clearly.
- Core business rules, including matching and assignment validation, can be tested independently.
- Code changes are developed in separate branches and reviewed by the other team member before merging into main.
- Meaningful work items have an assigned owner, priority, deadline, and acceptance criteria.
- Setup, configuration, and testing instructions are documented.
- Automated tests cover the main business rules and protected operations.

NFR-06. Compatibility and Responsiveness

Priority: P1

Requirement: The system shall provide a responsive web interface for common desktop and mobile browser sizes.

Acceptance criteria:

- Main pages remain usable at desktop, tablet, and mobile viewport sizes.
- Forms, event cards, schedules, and application lists do not require unintended horizontal scrolling at supported viewport sizes.
- Core workflows are tested in at least one current desktop browser and one mobile browser.
- The initial release does not require installing a native mobile application.

NFR-07. Privacy and Data Minimization

Priority: P0

Requirement: The system shall collect and display only the information needed for volunteer recruitment and event coordination.

Acceptance criteria:

- Registration and profile forms request only information justified by the documented requirements.
- Private profile information is accessible only to authorized users.
- Volunteer contact and profile details are not exposed through public event listings.
- The system does not expose personal data in unauthorized search results or API responses.
- Test cases verify access restrictions for personal information.

NFR-08. Testability and Evaluation

Priority: P0

Requirement: The system shall support reproducible verification of its core workflows and matching method.

Acceptance criteria:

- Automated tests cover registration validation, application status changes, matching rules, assignment capacity, schedule conflicts, permissions, and attendance.
- Integration tests verify interactions between the frontend, backend, and database where applicable.
- End-to-end tests cover at least one volunteer workflow and one organizer workflow.
- Matching evaluation uses a documented dataset, baseline, and evaluation procedure.
- Usability evaluation records its method, participants or test scenarios, and findings.
- Reported evaluation results are based on executed tests or collected evidence.

4. Requirement Prioritization Summary

Feature area	**Priority**	**Reason**
Registration and authentication	P0	Provides secure account access.
Volunteer profiles	P0	Supplies information for eligibility and matching.
Events and role creation	P0	Establishes the volunteering opportunities.
Applications and status tracking	P0	Supports the recruitment workflow.
Intelligent matching	P0	Represents a central research and implementation component.
Assignments	P0	Converts recruitment decisions into confirmed placements.

Scheduling and conflict prevention	P0	Prevents unsuitable or overlapping assignments.
Role-based permissions	P0	Protects data and administrative operations.
Security and data integrity	P0	Essential for trustworthy operation.
Task coordination	P1	Improves communication of responsibilities.
Notifications	P1	Keeps volunteers informed of changes.
Attendance tracking	P1	Supports participation monitoring and evaluation.
Search and filtering	P1	Makes opportunities easier to discover.
Performance and responsive interface	P1	Improves practical usability.
Advanced reporting	P2	Can extend the MVP after core workflows are complete.

5. Verification Plan

Each requirement will be verified using one or more of the following methods:

Verification method	**Purpose**	**Examples**
Unit testing	Verify individual business rules.	Matching scores, availability checks, role capacity.
Integration testing	Verify communication between components.	Authentication, database persistence, application updates.
End-to-end testing	Verify complete user workflows.	Volunteer applies; organizer approves; volunteer views assignment.
Permission testing	Verify access restrictions.	Volunteer attempts to edit another user's profile.
Performance testing	Measure response times under defined conditions.	Event listing and matching execution.
Usability evaluation	Assess ease of completing core tasks.	Finding an event and applying for a role.

Matching evaluation	Compare recommendations with a documented baseline and reviewed judgments.	Skill-based ranking and eligibility filtering.

A requirement will be considered complete when its acceptance criteria have been checked, the relevant evidence has been recorded, and any identified defects have been addressed or documented.

6. Scope and Assumptions

- The first release is a web-based MVP for Astana IT University.
- The main user groups are student volunteers, event organizers, and administrators.
- Matching recommendations support organizer decisions; they do not automatically replace organizer approval.
- The initial matching method will prioritize explainability and testability.
- The platform will use recorded availability and event schedules to identify conflicts.
- Native mobile applications and external university-system integrations are outside the initial scope.
- Advanced machine learning, automated email or SMS delivery, and advanced analytics may be considered for future versions.
- Requirements may be revised after discussions with student volunteers and event organizers. Any changes will be documented and reviewed by the team.

7. Definition of Done

A feature is considered complete when:

1. Its implementation satisfies the agreed requirements.
2. All applicable acceptance criteria pass.
3. Relevant automated tests are implemented and pass.
4. Access permissions and input validation are verified where applicable.
5. The code has been reviewed by the other team member and merged into main.
6. Relevant documentation is updated.
7. Known limitations and unresolved issues are recorded in the project repository.

Document status: Initial requirements draft. Priorities, acceptance criteria, and performance targets will be reviewed against stakeholder feedback, implementation constraints, and supervisor recommendations.